

RL Training Evidence

Purpose

This document records reproducible reinforcement-learning evidence for the Adaptive RL Gym Coach MVP. It is written for review-panel questions about machine learning, deep learning, and reinforcement learning.

The goal is evidence, not model marketing. The DQN result is reported exactly as measured. The reward function, training seeds, evaluation seeds, and metrics were fixed before running final experiments. No reward weights, seeds, or evaluation settings were changed after seeing results.

Scope Boundary

Topic	What this project proves	What it does not claim
RL training	DQN can be trained on a deterministic squat-coaching simulator and evaluated across fixed seeds.	It does not prove real human learning from long-term user feedback.
Real squat video	The real video validates pose extraction, rep segmentation, feature extraction, policy inference, explanations, overlay video, and dashboard path.	It is not used as RL training data.
Injury risk	Risk is a biomechanical proxy from torso lean, knee tracking, fatigue, and visible asymmetry.	It is not clinical diagnosis.
Explainability	Each action has state values, threshold triggers, model action, safety decision, reward, and evidence frame.	It is not Grad-CAM because no CNN posture classifier is trained.

Fixed Experiment Design

Training Seeds

13, 17, 23, 29, 31

Held-out Evaluation Seeds

1001 through 1030

Training Budget

50,000 timesteps per seed
5 training seeds
250,000 total DQN timesteps

State Vector

The public state vector stayed backward compatible with dashboard and tests.

Index	State value	Meaning
0	normalized knee angle	Squat depth and knee movement signal
1	normalized hip angle	Hip range signal
2	normalized torso angle	Forward lean signal
3	rep progress	Position inside simulated set
4	fatigue score	Estimated fatigue in current session
5	injury risk score	Estimated movement risk
6	repeated mistake count	Recurring form issue memory
7	previous action id	Last coaching action
8	skill level encoding	Beginner, intermediate, advanced context

Action Space

Action count stayed fixed at eight.

Action id	Action
0	verbal_cue
1	joint_highlight
2	slow_tempo
3	adjust_difficulty
4	recommend_rest
5	no_feedback
6	demonstration
7	breathing_cue

Simulator Dynamics

Each simulator step represents one squat repetition. The environment starts from seed-controlled fatigue, risk, movement quality, repeated mistakes, and skill level. Action choice affects next-step fatigue, risk, movement quality, and repeated mistake count.

The heuristic policy is also used as an interpretable reference policy:

Condition	Heuristic action
High fatigue	<code>recommend_rest</code>
High risk	<code>joint_highlight</code>
Repeated mistakes	<code>demonstration</code>
High torso lean	<code>slow_tempo</code>
Shallow pattern with repeated issue	<code>adjust_difficulty</code>
Moderate fatigue	<code>breathing_cue</code>
Single repeated issue	<code>verbal_cue</code>
Low-risk clean state	<code>no_feedback</code>

Reward Function

Reward combines improvement and penalties:

```
reward =  
    improvement in movement quality  
    + action-match reward  
    - fatigue penalty  
    - risk penalty  
    - repeated-mistake penalty  
    - correction penalty
```

The default reward config is `full_reward`. Ablations remove one penalty at a time:

Ablation	Removed component
<code>full_reward</code>	None
<code>no_fatigue_penalty</code>	Fatigue penalty
<code>no_risk_penalty</code>	Risk penalty
<code>no_repeated_mistake_penalty</code>	Repeated mistake penalty
<code>no_correction_penalty</code>	Correction penalty

Exact Commands

These commands were run from `/Users/tarry/Desktop/RL_gym_coach` inside `.venv`.

```
.venv/bin/python -m pytest -q  
.venv/bin/python -m src.train_rl --timesteps 50000 --seeds 13 17 23 29 31 --eval-seeds 1001  
.venv/bin/python -m src.evaluate --eval-seeds 1001 1002 1003 1004 1005 1006 1007 1008 1009 1
```

The exact executed command strings and environment metadata are also stored in:

`outputs/rl_training/run_manifest.json`

Runtime Environment

Item	Value
Git commit before this change set	71e6437
Python	3.12.14
Platform	macOS-26.5.1-arm64-arm-64bit
Stable-Baselines3	2.3.2
Gymnasium	0.29.1
NumPy	1.26.4
pandas	2.2.2
Matplotlib	3.9.2
PyTorch	2.14.0
OpenCV	4.10.0.84
MediaPipe	0.10.21
Streamlit	1.38.0

Generated Artifacts

Artifact	Meaning
outputs/rl_training/episode_rewards.csv	Replay SB3 Monitor episode rewards from simulator training. 20,830 rows.
outputs/rl_training/per_seed_eval_metrics.csv	Held-out metrics per DQN seed plus heuristic and random baselines.
outputs/rl_training/policy_comparison_metrics.csv	Model level comparison with winner labels.
outputs/rl_training/reward_ablation.csv	Reward ablation table.
outputs/rl_training/reward_curve.svg	Reward curve from logged training episodes.
outputs/rl_training/baseline_comparison.svg	DQN vs heuristic vs random reward chart.
outputs/rl_training/action_distribution.svg	Action frequencies on held-out simulator seeds.
outputs/rl_training/run_manifest.json	Commands, seeds, versions, platform, and manifest metadata.
models/rl_runs/coach_policy_seed_*.zip	Untrained DQN model per fixed seed.
models/coach_policy.zip	Canonical dashboard policy from first training seed, not best seed.

Policy Comparison

Lower is better for fatigue, risk, repeated mistake rate, and correction rate.
Higher is better for reward.

DQN standard deviation is computed across the five training-seed means. Heuristic and random standard deviations are computed across the 30 held-out evaluation episodes.

Metric	Direction	DQN mean	DQN std	Heuristic mean	Heuristic std	Random mean	Random std	Winner
Reward	max	0.714965	0.077572	0.871716	0.336870	-	2.742853	heuristic
Final fa-tigue	min	0.535816	0.004245	0.528991	0.025421	0.882249	10.769327	heuristic
Final risk	min	0.235068	0.003301	0.234774	0.071252	0.715206	0.162851	heuristic
Repeated mis-take rate	min	0.000556	0.001242	0.000000	0.000000	0.375000	0.195434	heuristic
Correction rate	min	0.440000	0.016736	0.427778	0.068276	0.897222	0.092027	heuristic

Result: DQN beats random on every metric, but the deterministic heuristic wins every measured metric on this simulator.

Per-seed DQN Results

Policy	Train seed	Eval episodes	Mean re-ward	Std re-ward	Mean final fa-tigue	Mean final risk	Mean repeated mis-take rate	Mean correction rate
DQN	13	30	0.654531	0.496267	0.540875	0.231814	0.000000	0.455556
DQN	17	30	0.640192	0.446013	0.539374	0.234180	0.000000	0.458333
DQN	23	30	0.769933	0.385033	0.534015	0.232862	0.000000	0.438889
DQN	29	30	0.820604	0.423847	0.530491	0.236341	0.000000	0.425000
DQN	31	30	0.689565	0.591240	0.534324	0.240141	0.002778	0.422222
Heuristic baseline		30	0.871716	0.336870	0.528991	0.234774	0.000000	0.427778
Random baseline		30	-	2.742853	0.882249	0.715206	0.375000	0.897222
				10.769327				

Reward Ablation

Ablation	Policy	Mean reward	Std reward	Eval episodes
full_reward	DQN	0.714965	0.472602	150
full_reward	heuristic	0.871716	0.336870	30
full_reward	random	-10.769327	2.742853	30
no_fatigue_penalty	DQN	1.710659	0.440999	150
no_fatigue_penalty	heuristic	1.861043	0.318853	30
no_fatigue_penalty	random	-9.365112	2.664681	30
no_risk_penalty	DQN	1.815331	0.340046	150
no_risk_penalty	heuristic	1.969429	0.169651	30
no_risk_penalty	random	-8.783970	2.429781	30
no_repeated_mistake_penalty	DQN	0.730965	0.449754	150
no_repeated_mistake_penalty	heuristic	0.871716	0.336870	30
no_repeated_mistake_penalty	random	-7.320327	1.373554	30
no_correction_penalty	DQN	1.005365	0.458146	150
no_correction_penalty	heuristic	1.154049	0.331344	30
no_correction_penalty	random	-10.177161	2.744822	30

Interpretation For Review Panel

Machine Learning

The project uses supervised-style feature engineering from pose landmarks, then policy learning over those features. No labeled real-world dataset is invented. Synthetic trajectories provide controlled states and rewards for RL proof.

Deep Learning

MediaPipe Pose supplies learned pose landmarks, but this project does not train MediaPipe. Stable-Baselines3 DQN uses a neural Q-network to estimate action values for the eight coaching actions.

Reinforcement Learning

The RL setup is a Markov Decision Process:

- State: 9 movement and profile values.
- Action: 8 coaching actions.
- Transition: simulator updates fatigue, risk, repeated mistakes, and movement quality.
- Reward: improvement minus safety and over-correction penalties.
- Policy: DQN trained for fixed timesteps across fixed seeds.

Honest Result

The heuristic policy won every metric on the current simulator. This is acceptable evidence because it shows DQN training was run and measured honestly,

not tuned to claim a false win. DQN still clearly improves over random, which supports that it learned useful action preferences, but it has not surpassed the hand-designed safety heuristic.

Reproducibility Checks

Run:

```
.venv/bin/python -m pytest -q
.venv/bin/python -m src.train_rl --timesteps 50000 --seeds 13 17 23 29 31 --eval-seeds 1001
.venv/bin/python -m src.evaluate --eval-seeds 1001 1002 1003 1004 1005 1006 1007 1008 1009 1
```

Expected verification:

- `outputs/rl_training/episode_rewards.csv` exists and contains real training episode rows.
- `outputs/rl_training/policy_comparison.csv` contains DQN, heuristic, and random comparisons.
- `outputs/rl_training/reward_ablation.csv` contains all five reward ablations.
- `outputs/rl_training/run_manifest.json` records seeds, commands, versions, and platform.
- `pytest` passes.